

Gastón Silva

Fullstack Developer | Tech Lead | GIS Systems

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PROFILE

Fullstack Developer and Systems Analyst focused on backend. Tech Lead of a 3-person team in the public sector, leading the development of a municipal geospatial (GIS) platform. My approach stands out for proactively resolving technical problems before they escalate: identifying unreported security vulnerabilities, fixing performance bottlenecks, and catching hard-to-detect bugs in systems already in production. Outside of work I keep the same habit, designing and shipping my own end-to-end projects, including an ERP (management system) for a food-industry business. Experience migrating legacy systems to modern architectures and leading technical teams.

EXPERIENCE

Fullstack Developer → Tech Lead — Municipality of Córdoba — Smart City & Digital Transformation Office
2024 – Present

Municipal Geoportal (current project, Tech Lead of a 3-developer team)

- Led the design and development of a municipal geoportal for creating and publishing interactive maps (Python/FastAPI + Vue 3/Quasar), used by multiple government areas: 190+ published layers and 103 REST endpoints across 14 modules.
- Designed a "catalog over the data" architecture: the system manages identity, permissions and organization for layers without duplicating the underlying geospatial data (GeoServer via WMS/WFS/WMTS), standardizing data access through stored procedures in PostgreSQL with versioned migrations over a live production environment.
- Optimized handling of large geographic datasets with vector tiles (MVT) and dynamic filtering, removing an N+1 pattern and adding in-memory caching plus server-side pagination on both frontend and backend.
- Implemented JWT authentication with federated SSO and role/work-group permissions; proactively identified and remediated unreported security vulnerabilities in the system.
- Built a dedicated accessibility mode ("Senior Citizens Portal") and integrated a routing engine (OSRM), geocoding, cloud storage, and multi-format export (Excel/CSV/KML/GeoJSON).
- Managed deployments (Docker) and database maintenance on production Linux (Ubuntu) servers.

Municipal Workforce Management System (MBTA) and Communications Portal — in parallel

- Rebuilt the backend and frontend of both systems simultaneously, migrating from PHP (Laravel) to FastAPI and Vue.js (Quasar) without discontinuing service during the transition.
- Integrated APIs and optimized queries on relational databases (PostgreSQL, SQL Server).

E-Procurement System

- Contributed to a mission-critical electronic procurement system for the city, ensuring stability and performance at government scale.
- Designed high-performance APIs with Python (FastAPI) for the backend and reactive interfaces with Vue.js (Quasar).

MOKKDEV — INDEPENDENT PROJECTS

Management System (ERP) — La Porteña S.R.L.

Client: family business (food industry) — actively growing project. Started as a thesis project; currently being refactored and deployed to real production outside the academic setting.

- Backend in Node.js/Express + Sequelize on PostgreSQL, covering the business end-to-end: purchasing, sales, production, raw materials, delivery notes, purchase/sales tax, expenses/income, clients and suppliers.
- JWT authentication, schema validation with Zod, PDF report generation and Excel export.
- Frontend in Vue 3/Quasar consuming the API via Axios, with data-entry forms per entity and data export.

Futbols — Amateur Football Management PWA

Independent project

- Vue 3/Quasar + Firebase/Firestore, with scheduled Cloud Functions (custom queues, per-minute jobs) for automatic notifications.

- Group system with roles and permissions, player statistics accumulated through idempotent functions (preventing duplicates or data manipulation), leaderboard, and match/venue management.

Panini World Cup 2026 Album — Collection PWA

Independent project

- Vue 3/Quasar + Firebase, real-time and offline sync of 980 items, with a trading view between users.

EDUCATION

Systems Analyst Degree

Colegio Universitario IES — 2022 – June 2025

B.Sc. in Computer Science (2nd year)

Universidad Nacional de Córdoba — 2019 – 2022

TECHNICAL SKILLS

Backend: Python (FastAPI), Node.js (Express), PHP (Laravel), C# (.NET Core, Blazor)

Frontend: Vue.js (Quasar), TypeScript

GIS: GeoServer, PostGIS, QGIS, Leaflet/Mapbox/OpenLayers

Databases: PostgreSQL, SQL Server, Firebase/Firestore

Infrastructure: Docker, Linux (Ubuntu), Git, GitHub

Architecture & practices: Legacy system migration, RESTful APIs, SOLID principles, Design Patterns

Leadership: Tech Lead of a 3-developer team (task assignment, code review, technical planning)

English: B1